SBT-DF205 – Case 2 Mobile and IoT Forensics

iOS Forensics Examination Report

Module:	SBT-DF205 Mobile and IoT Forensics — Module 2: iOS Forensics
Case reference:	SBT-DF205-C2
Case:	iPhone SE - iOS 13.4.1
Student:	Adekunle Ogunyemi
Registration number:	[Enter your registration number]
Environment:	Kali Linux
Evidence source:	Joshua Hickman's iOS 13.4.1 iPhone SE dataset, Digital Corpora
Time standard:	UTC, unless otherwise stated

Executive Summary

In this laboratory exercise, I carried out a forensic examination of a supplied iPhone SE dataset associated with iOS 13.4.1. I used Kali Linux to examine the available digital evidence, extract relevant records and compare information from different artefacts. My main objective was to understand what the evidence could establish about device setup, messages, attachments, Health activity and location-related events.

I began by checking the integrity of the supplied evidence archive. I used 7-Zip to test the ZIP archive, and the result was “Everything is Ok.” I also recorded its size and SHA-256 hash. I then worked with extracted files in a separate working directory so that my examination activities would not intentionally alter the original evidence.

During the examination, I analysed an Apple setup property list, the Messages database, recovered message attachments, the Health database and the location-related database. The setup property list recorded that setup had been completed and contained a SetupLastExit date of 21 March 2020. The Messages database contained records dated between 12 and 16 April 2020. Two messages referred to PDF documents that I was able to identify in the recovered attachment directory. I also identified six distinct workout records dated between 27 March and 3 April 2020.

The location database contained visit-related records with entry and exit times, coordinates, uncertainty values and data-point counts. I compared some of these records with the message timestamps. This showed that certain messages fell within a recorded location interval. However, I could not use this relationship alone to prove that the device remained continuously at one precise location or that a particular person sent the messages.

My examination identified three main relationships: messages and their associated PDF attachments, message timestamps that overlapped location-record intervals, and a sequence in which the recorded setup date preceded the later Health and message activity. I also considered alternative explanations, including the possibility that Health data had been synchronised from another device and that location records represented learned visits rather than continuous tracking.

Some parts of the assignment still require verified results, including account associations, contact and call-history analysis, Safari, Calendar, the Photos database and KnowledgeC activity. I have identified these as outstanding tasks rather than treating them as completed.

Overall, this practical helped me understand how to preserve evidence, examine iOS databases, convert timestamps and compare different artefact types. My conclusions are limited to the records I examined, and further corroboration is needed before making stronger claims about user identity, exact location or the circumstances surrounding individual events.

1. Introduction

For this practical, I examined a supplied iPhone SE forensic dataset associated with iOS 13.4.1. The exercise required me to investigate different iOS artefacts and use their contents to reconstruct a supported sequence of device-related activity.

I approached the examination by preserving the source evidence, creating a separate working area, identifying relevant databases and property lists, and extracting records using suitable command-line and database tools. I also considered the importance of interpreting timestamps correctly and distinguishing information directly stored in an artefact from conclusions that require further evidence.

1.1 Objectives

The objectives of my examination were to:

- Preserve the supplied evidence and record its integrity.
- Examine device setup information, settings and account-related artefacts.
- Extract messages and trace their associated attachments.
- Examine Safari history and Calendar records.
- Investigate photographs and media metadata.
- Examine KnowledgeC activity records.
- Analyse Health and location-related records.
- Construct a timeline of at least 12 events from four or more artefact families.
- Identify correlations, consider alternative interpretations and explain the limitations of the findings.

2. Evidence Handling and Methodology

2.1 Preserving the evidence

I began by identifying the supplied evidence archive and checking whether it could be read successfully. I used 7-Zip to test the archive. The result was “Everything is Ok,” indicating the archive passed the tool's integrity test.

The recorded archive details were:

Property	**Result**
Filename	ios_13_4_1.zip
File size	9,629,286,492 bytes
SHA-256	761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551
7-Zip test	Everything is Ok
Working directory	$HOME/SBT-DF205-Case-2-notes/

I kept my analysis material separate from the original evidence. I used extracted working files for database queries and metadata examination. This reduced the risk of unintentionally changing the original evidence during analysis.

The recorded hash above belongs to the ZIP archive. The instructor also requires recording the source TAR hash and rechecking it after examination. I must ensure that the TAR hash, any comparison results and the relevant processing details are included in the final handling log. I should not use the ZIP hash as if it were the TAR hash.

2.2 Organising the working files

I organised the examination material under the working directory: $HOME/SBT-DF205-Case-2-notes/

The main locations I used or examined included:

- task3/Volumes/JOSH/NoTar-13-4-1/ for extracted file-system material.
- task7/health/healthdb_secure.sqlite for Health records.
- task7/routined/Cache.sqlite for location-related records.
- task8/setup_evidence/ for setup property-list material.
- task8/recovered_attachments/ for recovered message attachments.
- task8/reports/ for examination output and reports.

I used 7-Zip for archive testing, SHA-256 hashing for integrity checks, Python to parse property-list data, SQLite queries to examine databases and timestamp conversion to express event times in UTC.

I recorded the source paths and relevant file information in my working notes. For the final submission, the evidence register and handling log must contain the precise source paths, evidence identifiers, tool versions and processing actions.

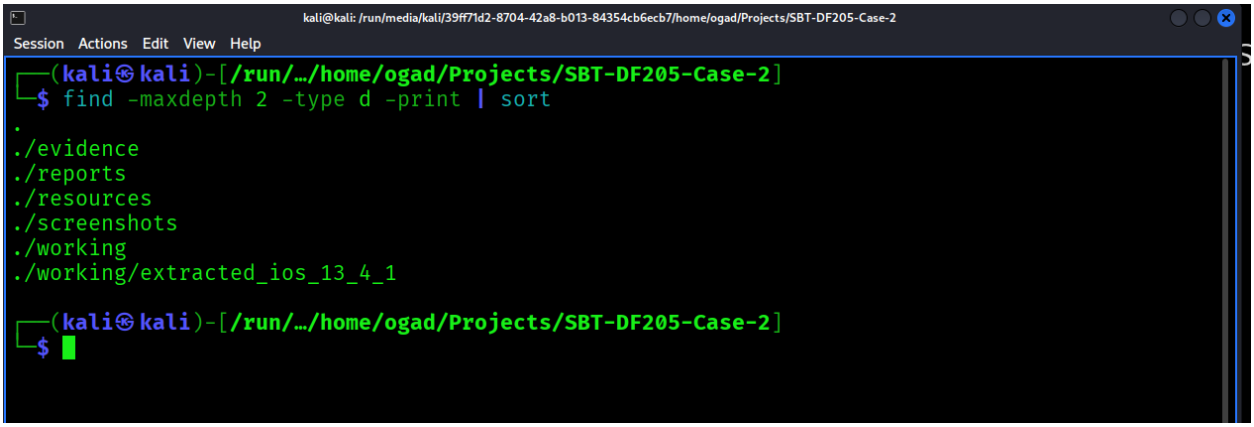


Figure_01_Case_Directory

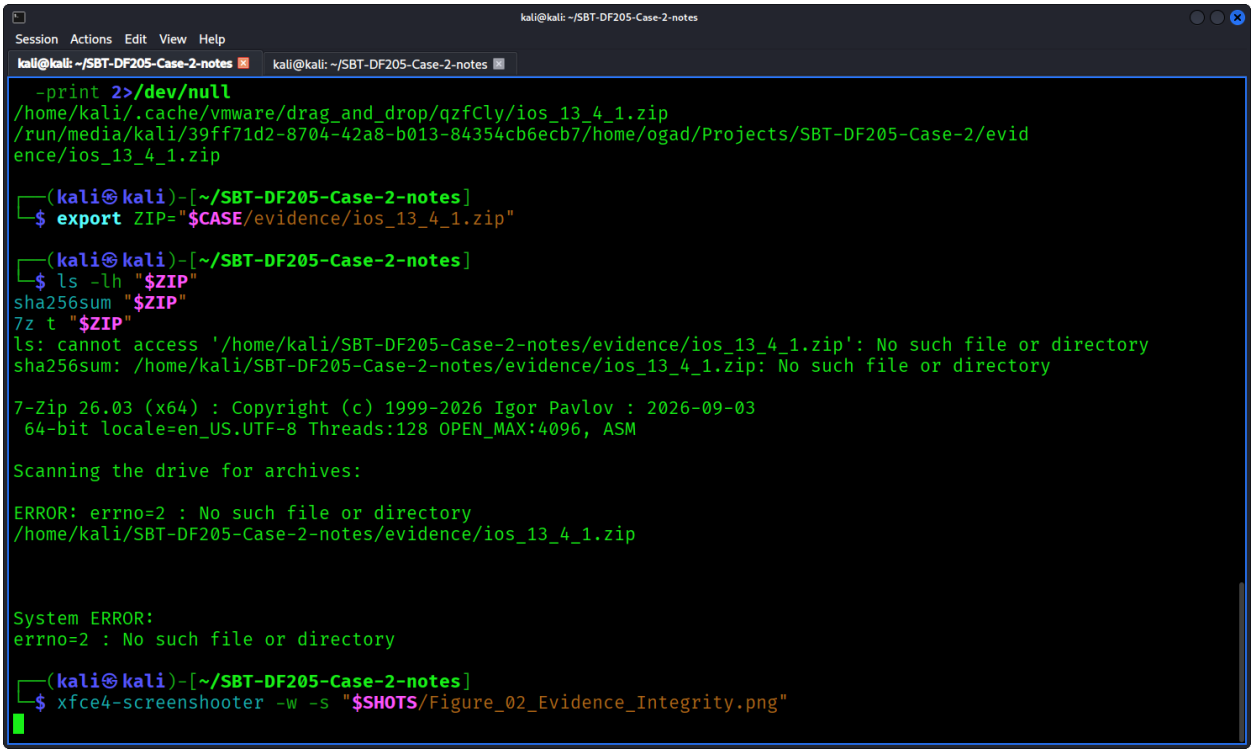


Figure_02_Evidence_Integrity

3. Device Information, Accounts and Settings

3.1 Examiningthe setup property list

I examined the following property list:
private/var/mobile/Library/Preferences/com.apple.purplebuddy.plist

The recorded SHA-256 hash was:
117247c482c236203929acce605ddf7389cb8b6b337f2662c87d7f7d5056515d

I also recorded the hash of com.apple.purplebuddy.notbackedup.plist:
b0246ab32fdd3716af03df269790f9cdcf6342a871021136fcb41306dbeb086

I parsed the main property list and identified the following values:

Property	Recorded value	My interpretation
SetupState	SetupUsingiTunes	The property list stores this setup-state value.
RestoreChoice	true	A setup choice is recorded, but the value does not identify the source of a backup.
SetupDone	true	Setup is recorded as completed.
SetupFinishedAllSteps	true	The property list records completion of the setup steps.

SetupLastExit	21 March 2020, 21:49:08.216927	The recorded setup exit date.
SetupVersion	11	The stored setup version.
Language	en-US	The stored language setting.
Locale	en_US	The stored locale.
GuessedCountry	US	The stored country guess, not proof of physical location or nationality.
Build version field	17D50	A build value stored in the examined property list.
Hardware model field	N69AP	A hardware-model identifier.

From these results, I concluded that the property list records a completed setup process and a setup exit date of 21 March 2020. However, I did not treat RestoreChoice = true as proof that a particular backup had been restored. The SetupState value was reported as stored in the file, rather than as conclusive proof of the complete setup method.

I also found a product-version value of 13.3.1 under chronicle.features. I did not use this single value to establish the installed iOS version at acquisition. The iOS version and build should be corroborated with a suitable system configuration or log artefact.

3.2 Account associations

The assignment requires me to examine Accounts3.sqlite and relevant property lists to identify account associations and corroborate at least one association using another source.

The examination notes available to me do not yet contain enough verified query results to support a final account-association finding. I have therefore not attributed a particular Apple account or other account to the device in this section.

To complete this task, I need to query the relevant account database, record the appropriate account fields and row identifiers, and compare at least one result with an independent artefact. I will include only the information necessary to support the finding.

3.3 Settings

The property list records language, locale and setup-state information. These values provide evidence of stored configuration settings, but they do not independently identify the person who configured the device.

The assignment also requires two supported settings, such as location-services or backup-related information. I need to verify those settings from the appropriate artefacts before claiming that this part of the assignment is complete.

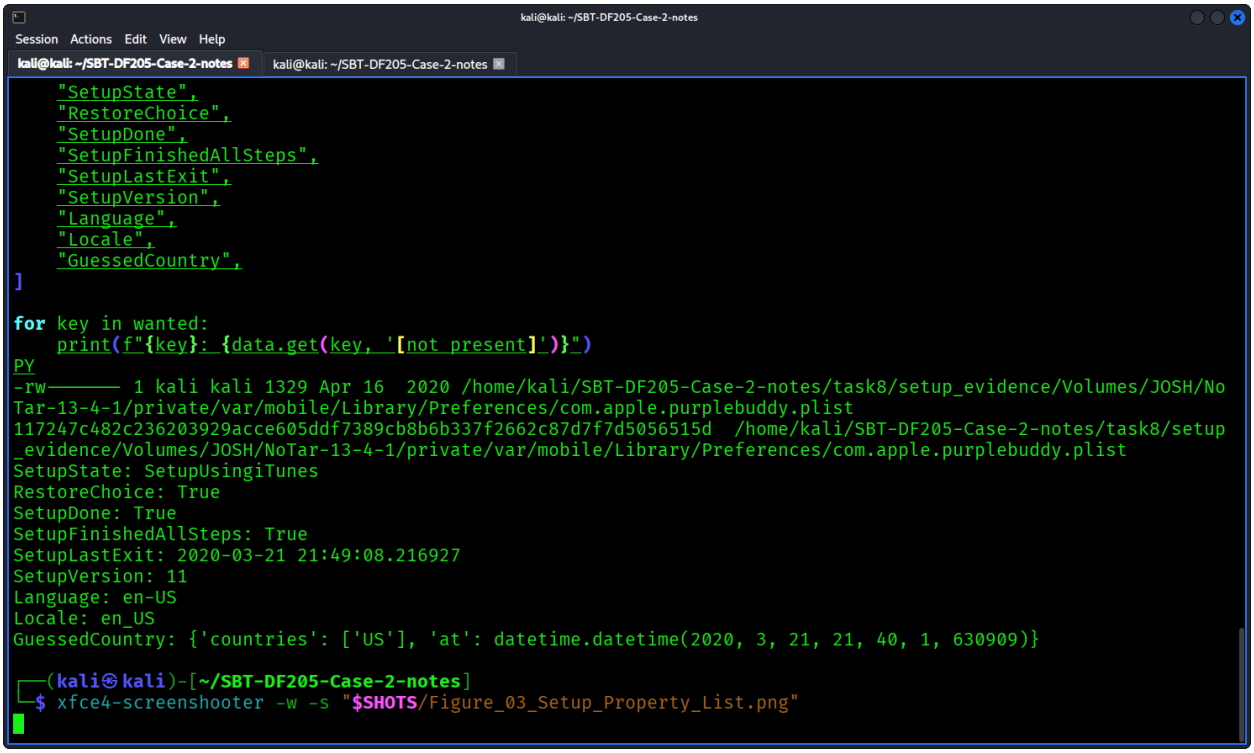


Figure_03_Setup_Property_List

4. Messages, Contacts, Calls and Attachments

4.1 Examining the Messages database

I examined the Messages database at:
private/var/mobile/Library/SMS/sms.db

For the message records listed below, I interpreted message.date as a timestamp stored in nanoseconds relative to Apple's epoch, beginning on 1 January 2001. I converted the values to UTC by dividing the raw value by one billion and adding the Apple epoch offset.

The conversion used in the SQLite query was equivalent to:
datetime(m.date / 1000000000 + 978307200, 'unixepoch')

The following records appeared in my examination output:

Message ID	UTC time	Direction	Recorded content or event
34	12 Apr 2020, 12:54:21	Outgoing	“Test iMessage. I hope this is a blue bubble!”
35	12 Apr 2020, 12:55:23	Incoming	Reply referring to generating a green bubble.
36	12 Apr 2020, 12:56:45	Outgoing	Message associated with IMG_0008.JPG.
37	12 Apr 2020, 12:58:01	Incoming	Message associated with FullSizeRender.jpeg.
38	12 Apr 2020, 13:16:19	Outgoing	“Here is Heather’s paper from DFIR Review.”
39	12 Apr 2020, 13:21:15	Incoming	“Here is the sysdiagnose paper. Enjoy!”
40	12 Apr 2020, 16:14:26	Incoming	Message from a phone-number handle referring to a green message.
41	12 Apr 2020, 16:15:00	Outgoing	Reply confirming the green message.
42	12 Apr 2020, 16:16:15	Incoming	Further message from the same phone-number handle.
43	15 Apr 2020, 16:20:41	Outgoing	“This is a test message for Siri and CarPlay”.
44	15 Apr 2020, 17:19:51	Incoming	Facebook friend-suggestion notification from handle 32665.
45	16 Apr 2020, 11:45:50	Outgoing	Message containing an iCloud Drive link.
46	16 Apr 2020, 12:17:58	Outgoing	“This is a test message for Siri dictation”.
47	16 Apr 2020, 12:56:49	Incoming	Reply referring to Siri dictation.
48	16 Apr 2020, 12:59:29	Incoming	Second reply referring to Siri dictation.

These results show that the database contains incoming and outgoing messages from 12 to 16 April 2020. The messages include ordinary conversation, attachment-related messages and references to Siri and CarPlay testing.

I treated the stored message text as evidence of what the database records. I did not use it alone to prove the real-world identity of the sender or recipient.

For the final submission, I need to preserve the full query output for at least five messages, including raw timestamps, converted UTC times, participant or handle, direction, service type and message text.

4.2 Recovering and examining attachments

I examined the recovered attachment directory:

$HOME/SBT-DF205-Case-2-notes/task8/recovered_attachments/

Two PDF attachments were consistent with the content of messages 38 and 39.

Message ID	Message content	Recovered file
38	Refers to Heather's paper from DFIR Review	How was an iPhone set up?.pdf
39	Refers to a sysdiagnose paper	iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf

I recorded the following file details:

Filename	Size	SHA-256
How was an iPhone set up?.pdf	846,528 bytes	82118655004fa94dc9257988cab0e02b7efda5fada929cbd92d04def11a136b6
iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf	7,073,725 bytes	255ac0cac61a3f701852bdd76a31f39e95b3760f04d53eb9e10a45cc2464cb3e

The first PDF was reported as eight pages, while the second was reported as 74 pages.

I also identified two image attachments:

Filename	Dimensions	Size	SHA-256
IMG_0008.JPG	1125 × 1156	676,542 bytes	df817f1c02f2c57c749a6a2d86e6beaafeb2186703745650c6bd784075757edf
FullSizeRender.jpeg	2212 × 1306	368,536 bytes	1eb7fcc25d433fe5bf2783fec140c0088b2fae117bc15473ce7c6ca1e1827b4d

The message text and the recovered PDFs were consistent with the documents being associated with the corresponding messages. This is a useful relationship between two types of evidence.

The recorded file modification times were not automatically treated as original capture times. A file's modification time can reflect copying, extraction or another process, so I would need supporting metadata before making a claim about when an image was captured.

4.3 Contacts and call history

The assignment also requires me to examine AddressBook.sqlitedb, match at least two message or call identifiers to stored contacts where possible, and inspect at least three records from CallHistory.storedata.

I do not yet have verified query results for these databases in the examination notes. I have therefore left this part outstanding instead of assigning a contact name to a phone number or inventing call records.

To complete the task, I need to extract the relevant contact and call records, identify any supported matches and document the source fields used to make each match.

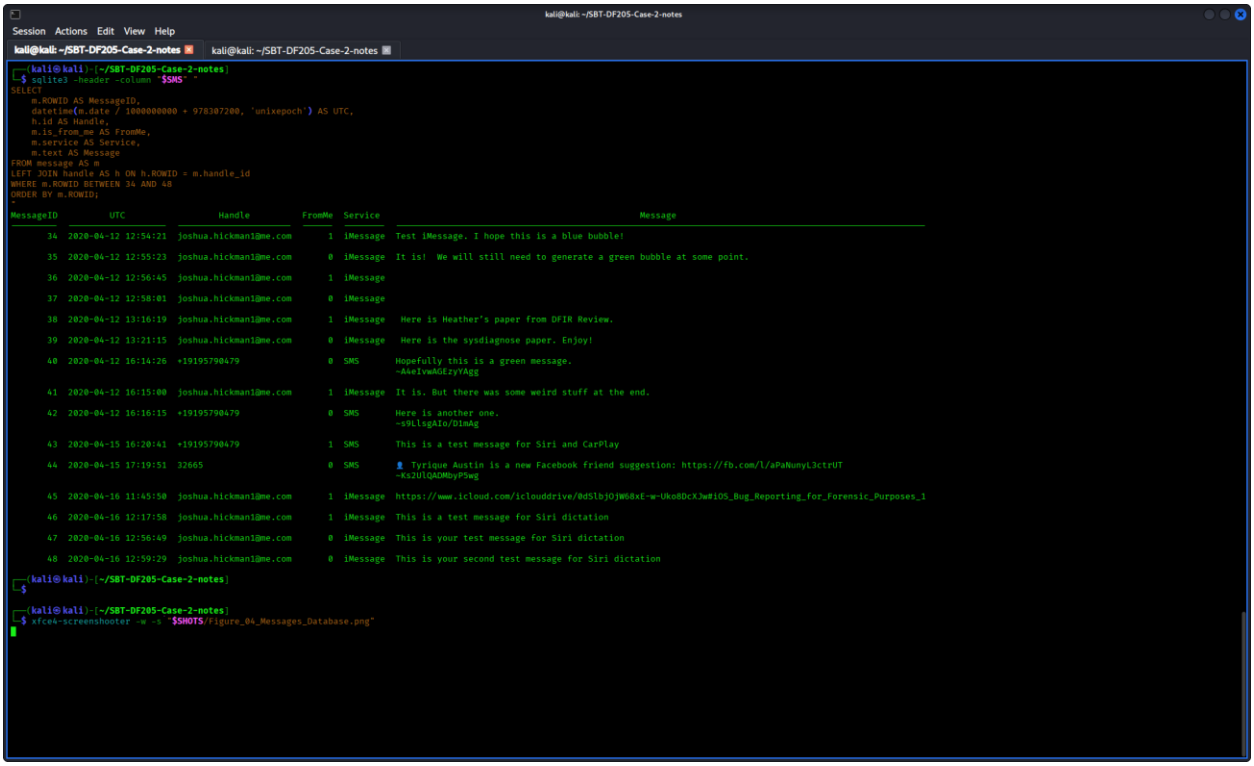


Figure_04_Messages_Database

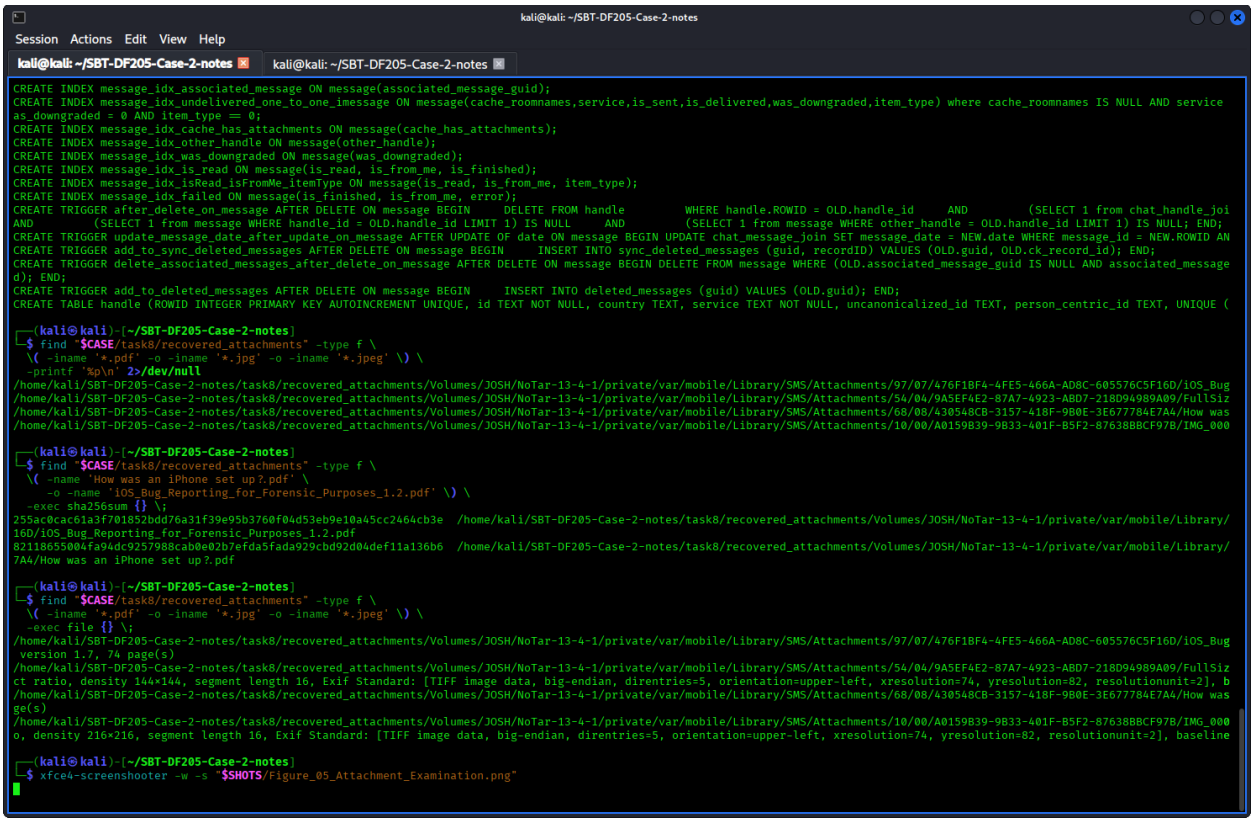


Figure_05_Attachment_Examination

5. Safari and Calendar Examination

The assignment requires an examination of Safari's History.db, including at least five visit records, and a review of Calendar.sqlitedb and Extras.db where those files are available.

I do not yet have a verified list of Safari visits or a completed Calendar examination in my current results. I have therefore not included specific URLs, appointments or browsing events as established findings.

To finish this part of the lab, I need to:

- ❖ Locate Safari's History.db in the extracted dataset.
- ❖ Review the database schema and extract at least five actual visit records.
- ❖ Record the URL, title where available, raw timestamp, timestamp field, epoch and unit.
- ❖ Convert the timestamp to UTC only after confirming the field's time format.
- ❖ Distinguish browsing history from bookmarks, open tabs and synced tabs.
- ❖ Inspect the relevant Calendar databases and document the tables and event records available.

If a database is absent or a query returns no relevant records, I will record the result and explain the limitation rather than inventing activity.

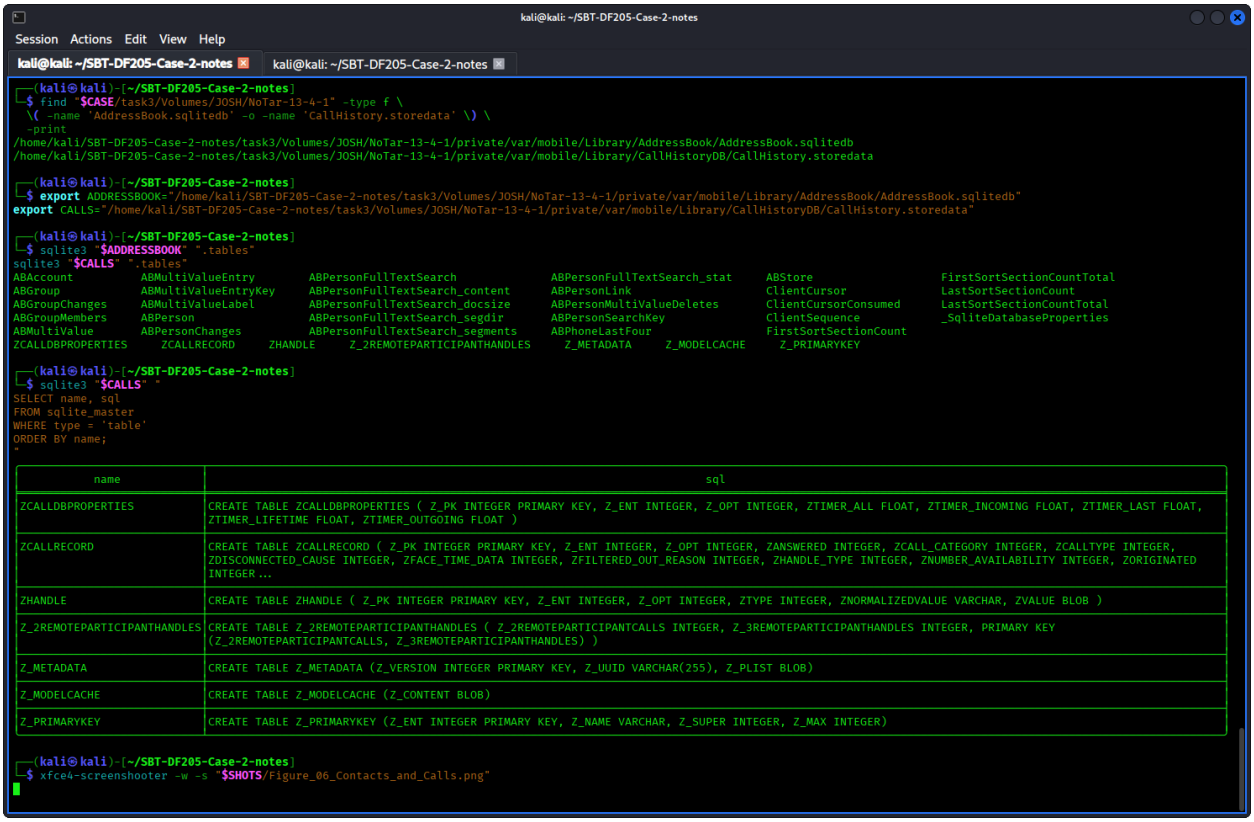


Figure_06_Contacts_and_Calls

6. Photographs and Media

The assignment requires examination of Photos.sqlite, including ZGENERICASSET and related metadata tables where available. At least three media records must be examined, including relevant filenames, paths, hashes, times and GPS information where recorded.

I have verified the hashes and dimensions of two image files recovered as Messages attachments. However, I have not yet established that these files complete the separate Photos database requirement.

To complete the task, I need to:

- ❖ Locate Photos.sqlite and identify the relevant tables.
- ❖ Select at least three media records.
- ❖ Record the database identifiers and available metadata.
- ❖ Locate the corresponding files where possible and calculate their SHA-256 hashes.

- ❖ Examine capture-related and added timestamps separately.
- ❖ Record time-zone offsets and GPS fields only where they are actually available.
- ❖ Compare at least one database record with the metadata of its corresponding file.

I will not treat a sharing or deletion indicator as proof that a complete transfer occurred or that a user deliberately deleted a file unless additional evidence supports that interpretation.

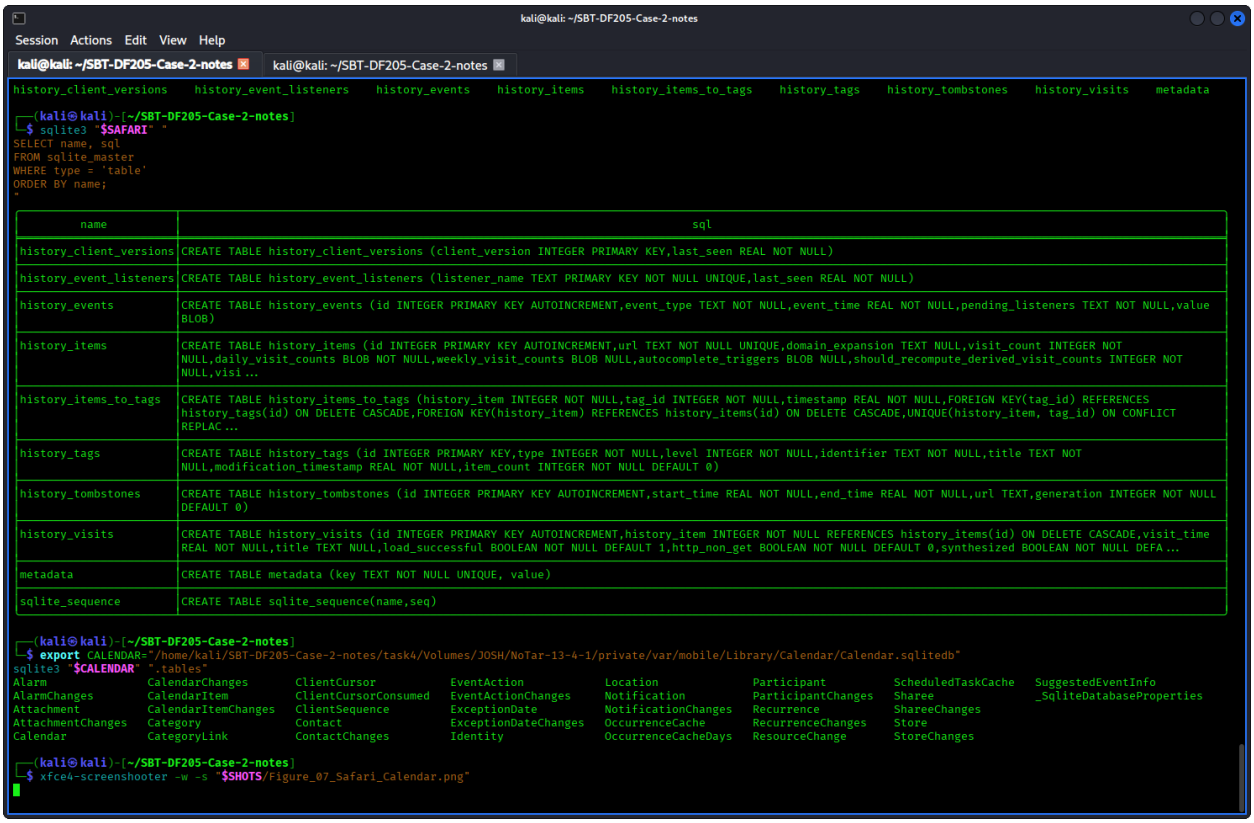


Figure_07_Safari_Calendar

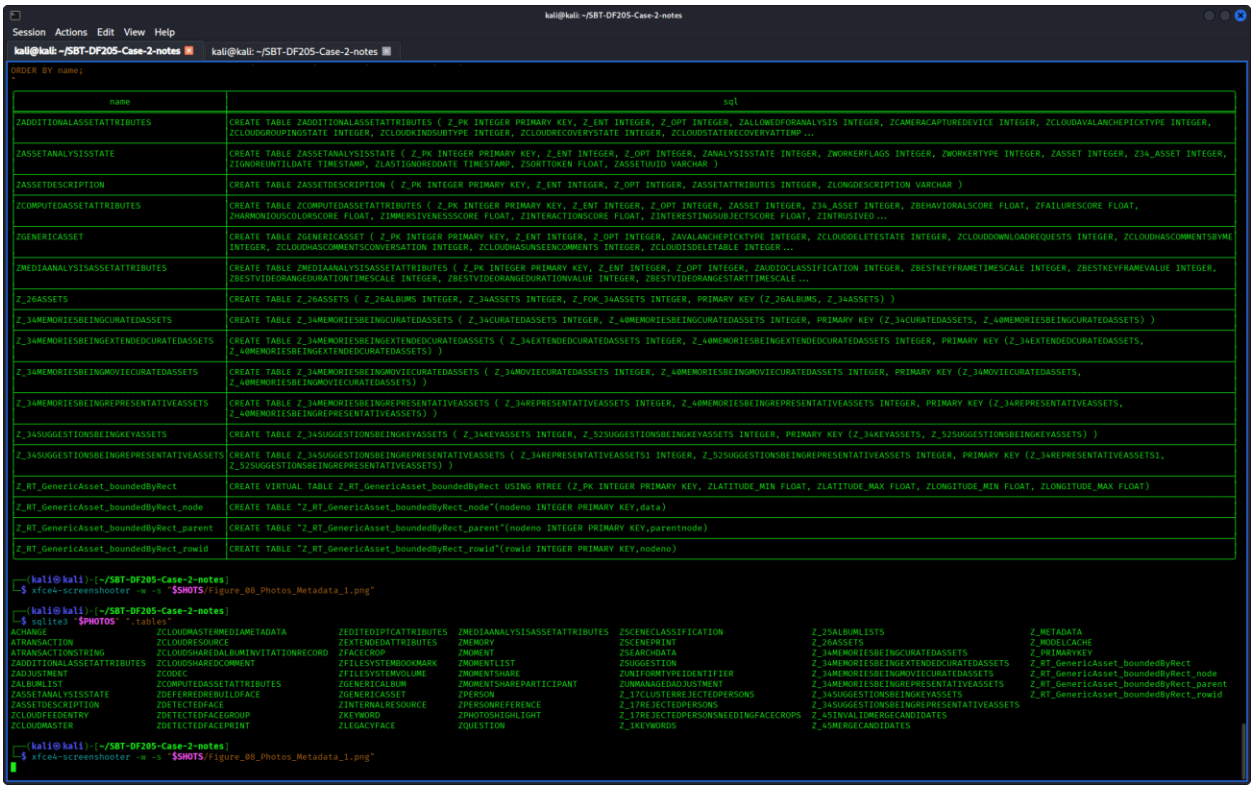


Figure_08_Photos_Metadata_1

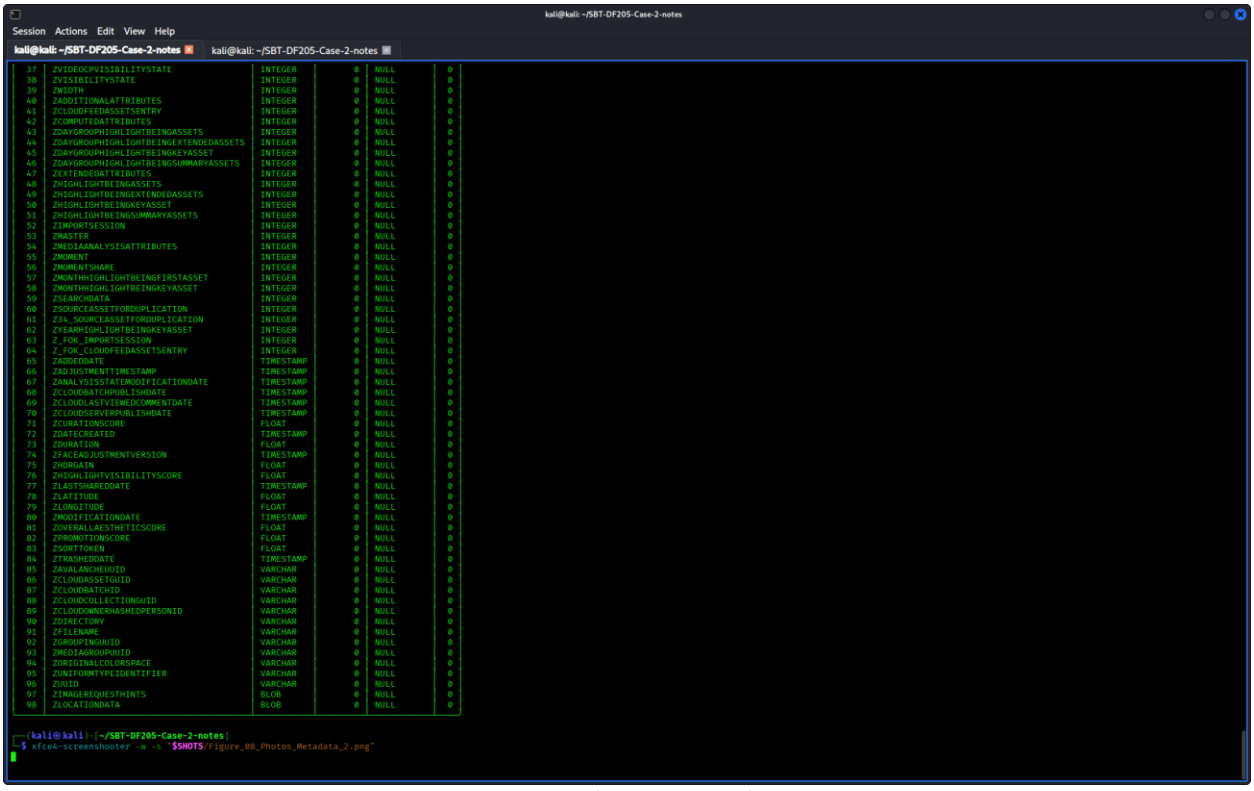


Figure_08_Photos_Metadata_2

7. KnowledgeC Activity

I also need to examine:
private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db

The assignment requires at least five relevant records from ZOBJECT and related metadata, such as application-focus or device-lock activity. The required fields include the stream name, record ID, associated application or value, raw timestamps, converted UTC times and duration where meaningful.

The current examination notes do not contain a verified five-record KnowledgeC extract. I have therefore not claimed that a specific application was active or that the device was locked or unlocked at a particular time.

To complete this task, I need to inspect the database schema, identify the appropriate timestamp fields, validate the epoch and unit, and extract the required records. I will then compare at least one activity interval with a message, browsing or media event and explain whether the intervals overlap.

An overlap can support a comparison between records, but it does not independently prove that the same person performed both activities.

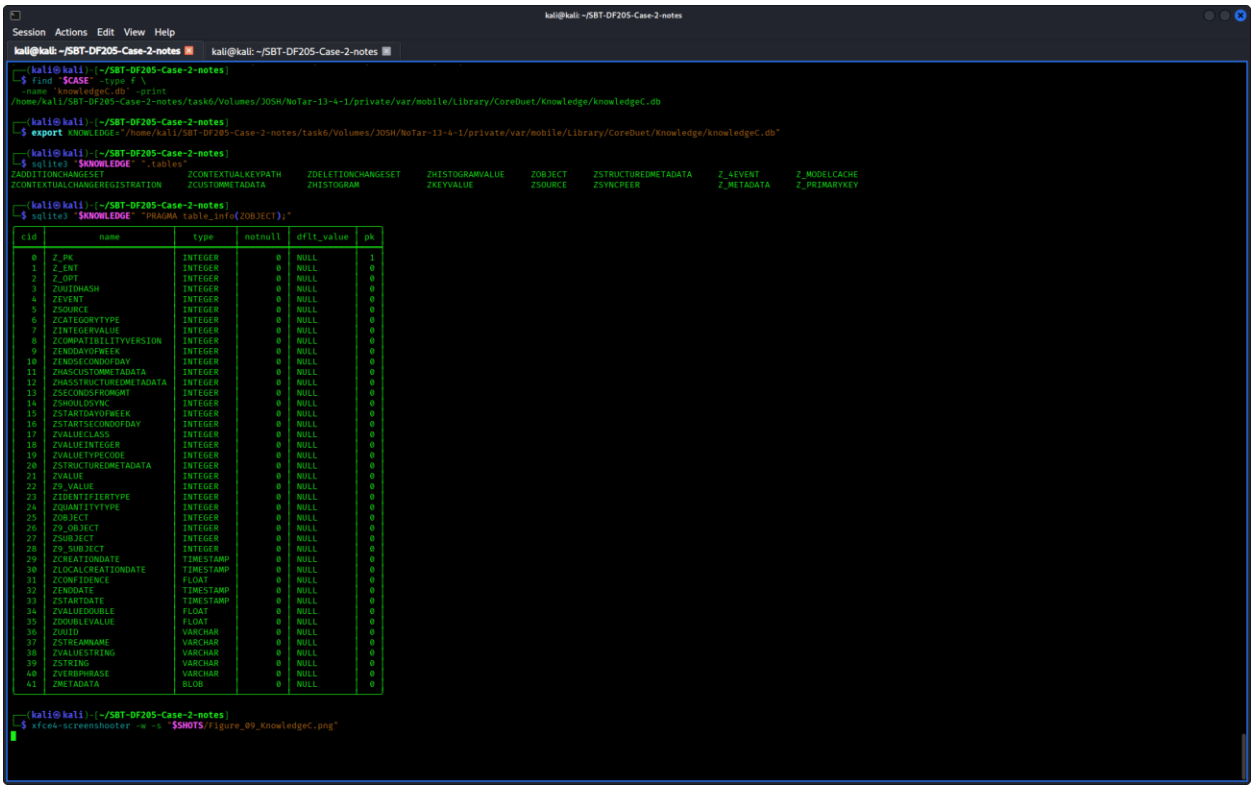


Figure_09_KnowledgeC

8. Health and Location Evidence

8.1 Health database

I examined the Health database at:
$HOME/SBT-DF205-Case-2-notes/task7/health/healthdb_secure.sqlite

I reviewed the workout_events and workouts tables. The available fields include workout event dates, types, durations, metadata and session identifiers, along with workout duration, energy, distance and activity-type values.

My query identified six distinct workout records:

Workout ID	First recorded event (UTC)	Last recorded event (UTC)	Duration recorded	Distance value	Energy value
15017	27 Mar 2020, 18:57:39	27 Mar 2020, 19:15:26	1,080.12 s	3.24798	208.57
18146	28 Mar 2020, 14:41:25	28 Mar 2020, 14:58:38	1,039.28 s	3.23536	203.11
24649	30 Mar 2020, 12:44:04	30 Mar 2020, 13:05:59	1,591.07 s	4.84550	302.58

30219	31 Mar 2020, 18:58:47	31 Mar 2020, 19:24:48	1,572.25 s	4.86485	302.86
36805	2 Apr 2020, 17:36:41	2 Apr 2020, 18:02:24	1,550.33 s	4.85304	301.89
44801	3 Apr 2020, 19:11:28	3 Apr 2020, 19:37:41	1,580.99 s	4.86275	369.99

The timestamps were interpreted as seconds relative to Apple's epoch and converted to UTC. I retained the numeric activity-type value rather than assigning it a name because I have not verified the mapping for activity code 37. I also need to confirm the units for the distance and energy fields before describing them more specifically.

These records establish that the examined database contains six workout entries with the listed times and stored values. They do not, on their own, establish who performed the activity. The data could have been synchronised from an Apple Watch or another source, so I would need source-device information and corroborating records before attributing the activity to a particular person.

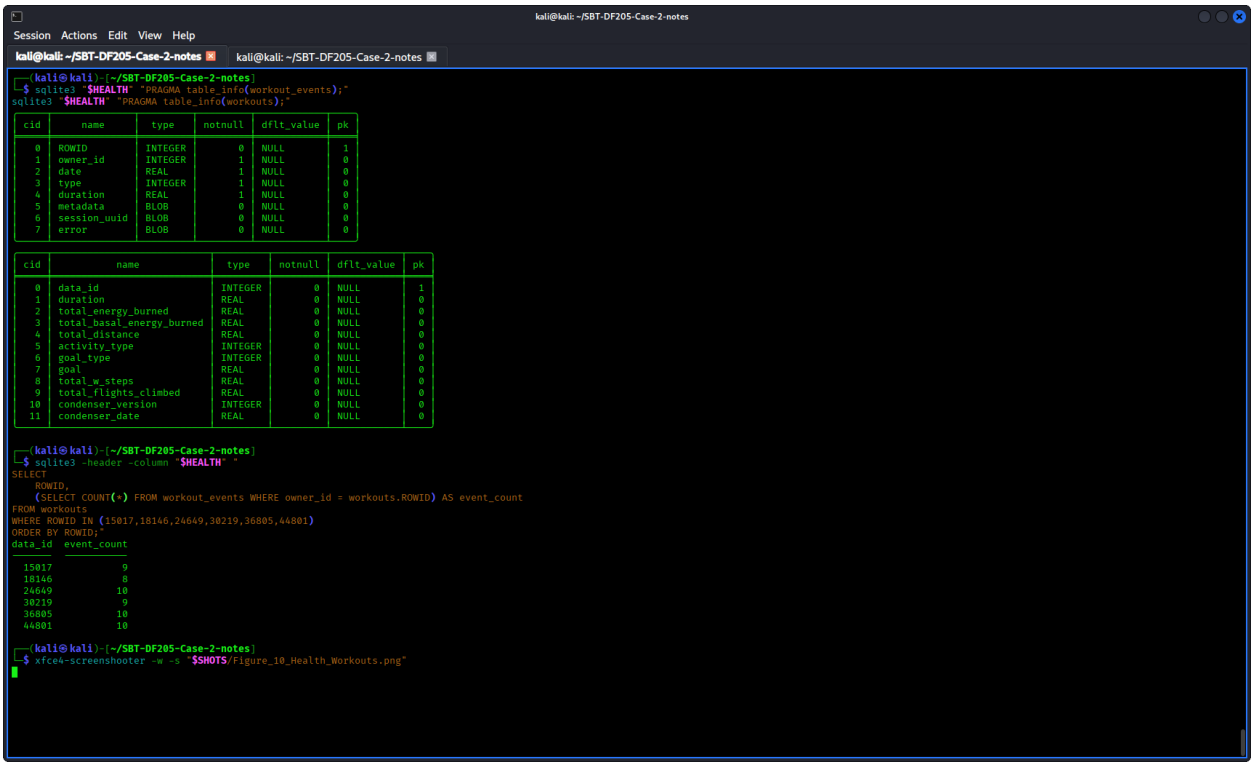


Figure_10_Health_Workouts

8.2 Location database

I examined the location-related database at:
$HOME/SBT-DF205-Case-2-notes/task7/routined/Cache.sqlite

The ZRTVISITMO table contains fields for visit entry and exit times, detection dates, coordinates, uncertainty, confidence and data-point counts.

I interpreted the timestamps in the selected records as seconds relative to Apple's epoch. The following records were identified:

Record ID	Entry (UTC)	Exit (UTC)	Approximate coordinates	Uncertainty	Data points
68	11 Apr 2020, 19:09:45	12 Apr 2020, 13:55:11	35.659603, -78.872855	32.78 m	1,220
70	12 Apr 2020, 13:55:12	12 Apr 2020, 13:56:00	35.659450, -78.876374	128.57 m	49
82	12 Apr 2020, 14:27:08	13 Apr 2020, 19:20:24	35.659297, -78.873080	24.08 m	5,492
83	12 Apr 2020, 14:27:08	14 Apr 2020, 12:38:07	35.659317, -78.873105	25.69 m	7,702
84	12 Apr 2020, 14:27:08	15 Apr 2020, 16:02:36	35.659341, -78.873066	28.56 m	14,977

I also observed that record 85 duplicates the entry and exit times, coordinates and data-point count of record 84, although the confidence value differs. I did not count record 85 as an independent visit without further evidence.

These records show that the database contains visit-related information with coordinates and uncertainty values. However, I did not interpret the long intervals as proof that the device remained continuously at a precise location. The meaning of the interval must be considered alongside the database's semantics and other evidence.

I also compared the selected location records with the Messages timestamps. Messages 40 and 41, recorded at 16:14:26 and 16:15:00 UTC on 12 April, fall within the long interval associated with records 82–85. This is a temporal overlap, not proof that a particular person was at the recorded coordinates when the messages were sent.

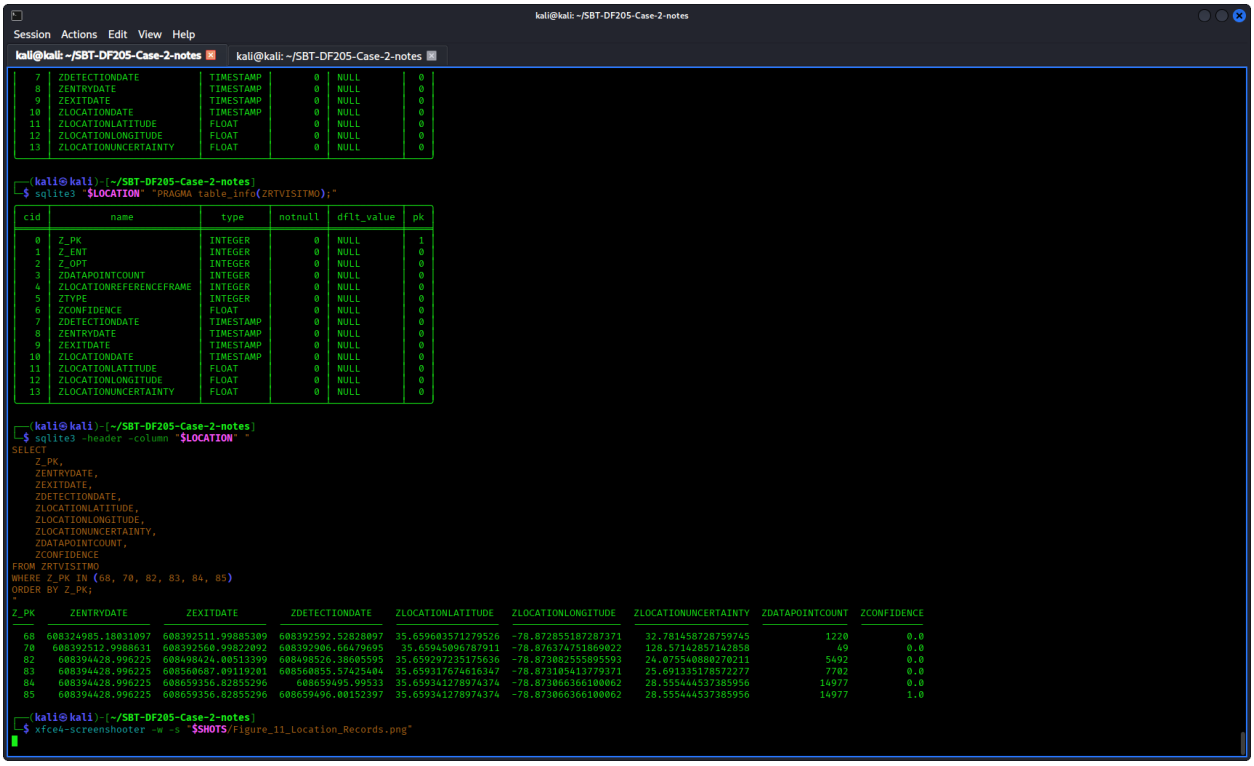


Figure_11_Location_Records

9. Timeline of Events

I used the available setup, Health, Messages and location records to prepare a timeline. This provides more than the required 12 events across four artefact families. The raw timestamps and their associated fields are retained so that the conversions can be checked.

Event	Raw time and field	Epoch and unit	UTC conversion	Description	Evidence ID
1	SetupLastExit = 2020-03-21 21:49:08.216927	Property-list date	21 Mar 2020, 21:49:08	Recorded setup exit date	SET-001
2	607028259.36280501, first workout event, ID 15017	Apple epoch, seconds	27 Mar 2020, 18:57:39	First recorded event for workout 15017	HLT-15017
3	607099285.47888005, first workout event, ID 18146	Apple epoch, seconds	28 Mar 2020, 14:41:25	First recorded event for workout 18146	HLT-18146
4	607265044.34228301, first workout event, ID 24649	Apple epoch, seconds	30 Mar 2020, 12:44:04	First recorded event for workout 24649	HLT-24649
5	607373927.61740398, first workout event, ID 30219	Apple epoch, seconds	31 Mar 2020, 18:58:47	First recorded event for workout 30219	HLT-30219

6	607541801.45553601, first workout event, ID 36805	Apple epoch, seconds	2 Apr 2020, 17:36:41	First recorded event for workout 36805	HLT-36805
7	607633888.75596297, first workout event, ID 44801	Apple epoch, seconds	3 Apr 2020, 19:11:28	First recorded event for workout 44801	HLT-44801
8	608388861591000064, message.date, ID 34	Apple epoch, nanoseconds	12 Apr 2020, 12:54:21	Outgoing iMessage test	MSG-034
9	608389005092000000, message.date, ID 36	Apple epoch, nanoseconds	12 Apr 2020, 12:56:45	Outgoing message associated with an image attachment	MSG-036
10	608390179362000000, message.date, ID 38	Apple epoch, nanoseconds	12 Apr 2020, 13:16:19	Outgoing message referring to a PDF	MSG-038
11	608390475684724608, message.date, ID 39	Apple epoch, nanoseconds	12 Apr 2020, 13:21:15	Incoming message referring to a sysdiagnose paper	MSG-039
12	608392512.9988631, ZENTRYDATE, location record 70	Apple epoch, seconds	12 Apr 2020, 13:55:12	Entry time for a short visit-related record	LOC-070
13	608400866607614080, message.date, ID 40	Apple epoch, nanoseconds	12 Apr 2020, 16:14:26	Incoming message from a phone-number handle	MSG-040
14	608400900444000000, message.date, ID 41	Apple epoch, nanoseconds	12 Apr 2020, 16:15:00	Outgoing reply	MSG-041
15	608659356.82855296, ZEXITDATE, location record 84	Apple epoch, seconds	15 Apr 2020, 16:02:36	Exit time stored for a long location interval	LOC-084
16	608660441673863040, message.date, ID 43	Apple epoch, nanoseconds	15 Apr 2020, 16:20:41	Outgoing message referring to Siri and CarPlay testing	MSG-043
17	608730350348999936, message.date, ID 45	Apple epoch, nanoseconds	16 Apr 2020, 11:45:50	Outgoing message containing an iCloud Drive link	MSG-045
18	608732278492000128, message.date, ID 46	Apple epoch, nanoseconds	16 Apr 2020, 12:17:58	Outgoing message referring to Siri dictation testing	MSG-046

I will preserve the complete timeline in Timeline.csv, including the exact raw value, source field, epoch and unit, converted UTC time, event description, source path and evidence ID.

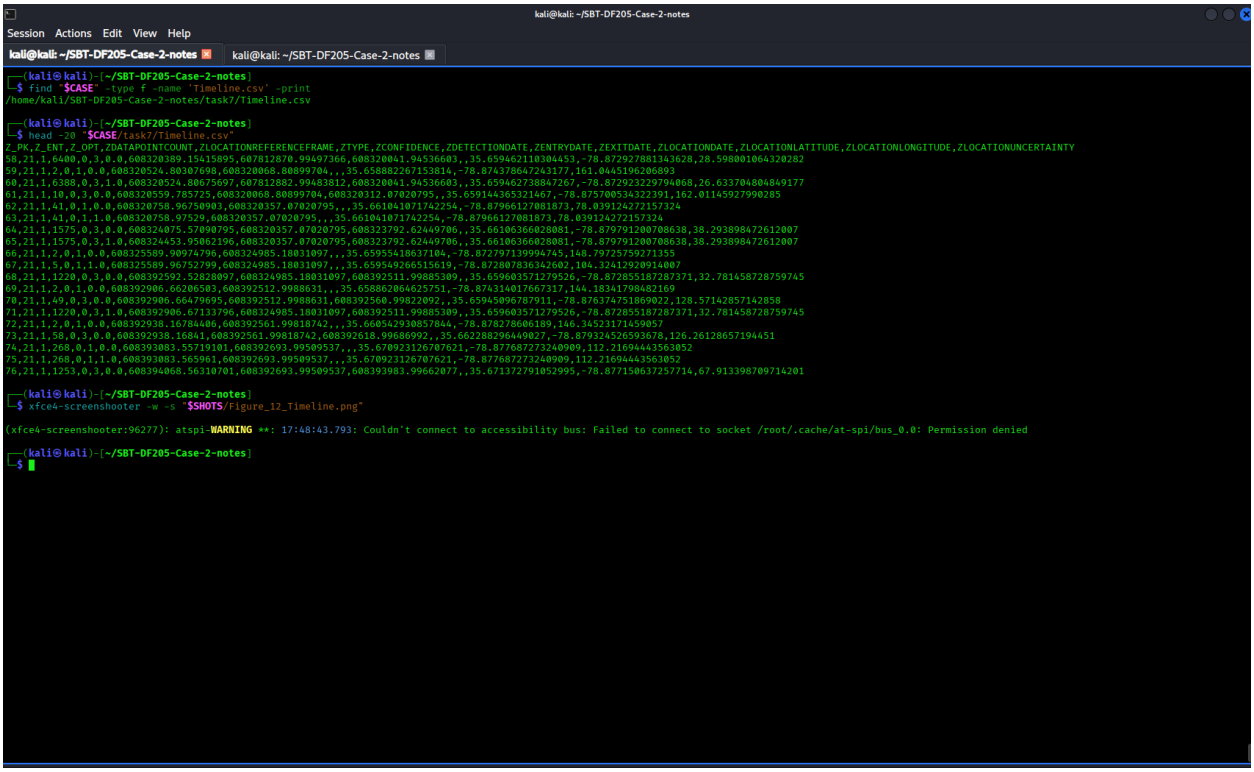


Figure_12_Timeline

10. Correlations and Alternative Interpretations

10.1 Correlation between messages and PDF attachments

I compared messages 38 and 39 with the recovered attachment files. Message 38 refers to Heather's paper from DFIR Review, while the recovered attachment is named How was an iPhone set up?.pdf. Message 39 refers to a sysdiagnose paper, while the recovered attachment is named iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf.

The message text and the recovered files are consistent with the PDFs being associated with the respective messages.

This does not prove that the recipient opened or read either document, nor does it establish who physically operated either device. Additional attachment database records and endpoint evidence would help establish the relationship more fully.

10.2 Correlation between messages and location records

I compared the timestamps of messages 40 and 41 with the location intervals in records 82–85. The messages were recorded at 16:14:26 and 16:15:00 UTC on 12 April

2020, while the long location-related interval began at 14:27:08 UTC on the same date.

The timestamps overlap. This supports a temporal comparison between the message records and the location records.

However, the interval does not prove that the device remained at one exact coordinate throughout that period. The uncertainty values and the meaning of the location records must be considered. Further location evidence and independent records would be required to make a stronger statement about the device's physical position.

10.3 Correlation between setup and later activity

The setup property list records a setup exit date of 21 March 2020. The Health database contains workout records from 27 March to 3 April, and the Messages database contains records from 12 to 16 April.

This establishes a chronological sequence in which the recorded setup date precedes the examined Health and message activity. It does not prove that the same person performed every later action or that the setup record identifies a specific backup source.

Additional system logs, corroborated account information and independent activity records would be required to strengthen this conclusion.

10.4 Alternative interpretation: device setup

One possible interpretation is that the setup property list reflects a setup or restoration process involving iTunes, based on SetupState = SetupUsingiTunes and RestoreChoice = true.

An alternative interpretation is that these values record a setup state and choice without establishing the exact restoration method or backup source. I therefore reported the stored values without treating them as conclusive proof of a particular backup being restored.

Backup metadata, setup logs and other configuration artefacts could help resolve this uncertainty.

10.5 Alternative interpretation: Health and location activity

One possible interpretation is that the Health and location records represent activity associated with the examined iPhone during the stated periods.

An alternative interpretation is that the Health data was synchronised from a wearable or another source and that the location records represent learned visits or aggregated estimates rather than continuous, precise tracking.

The current records do not independently identify the person who performed each activity. Health source information, additional location databases and independent corroborating evidence would help resolve these uncertainties.

11. Limitations and Outstanding Examination Tasks

During the examination, I identified areas where additional verified results are needed to meet all the assignment requirements.

Task	Current position	Further work required
Evidence preservation	ZIP hash and archive test recorded	Record source TAR hash, pre- and post-examination checks, receipt details and tool versions
Device and accounts	Setup property list examined	Corroborate device model, installed iOS version/build, account associations and one association using an independent source
Messages and attachments	Message records and four attachment files identified	Complete contact matching, attachment database linkage and examination of at least three call records
Safari and Calendar	Results not yet verified	Extract at least five Safari visits and inspect Calendar and Extras tables
Photos and media	Two image attachments have recorded hashes and dimensions	Examine Photos.sqlite and at least three media records, including available metadata
KnowledgeC	Results not yet verified	Extract at least five records and compare one interval with another artefact
Health and location	Six workout records and multiple visit-related records identified	Confirm units, timestamp conversions, source-device information and relevant additional location records
Timeline	Timeline drafted from four artefact families	Validate conversions and complete Timeline.csv
Supporting files	Findings recorded in examination notes	Complete Evidence_Register.csv, Queries.sql and Handling_Log.csv

I have not treated an unexamined database as evidence of an absence of activity. If a source is unavailable or a query returns no relevant records, I will document the attempt, output and limitation. This is necessary to distinguish a genuine negative result from a task that has not yet been completed.

12. Conclusion

In this practical, I examined a supplied iPhone SE forensic dataset using Kali Linux. I tested the supplied ZIP archive, recorded its SHA-256 hash and worked with extracted files in a separate working directory. I then examined setup information, message records, recovered attachments, Health data and location-related records.

The setup property list recorded a completed setup state and an exit date of 21 March 2020. The Messages database contained incoming and outgoing messages dated between 12 and 16 April 2020. I identified two PDF attachments whose filenames and contents were consistent with the messages referring to them. I also identified six workout records dated between 27 March and 3 April 2020 and location-related records containing timestamps, coordinates, uncertainty values and data-point counts.

By comparing the artefacts, I identified a relationship between message text and recovered PDFs, a temporal overlap between messages and a location-record interval, and a chronology in which the recorded setup date preceded later Health and message activity.

These findings provide a useful basis for reconstructing device-related activity. However, they do not independently prove the identity of the person responsible for every event, the exact location of the device at each message time or the source of every Health record.

I still need to complete and verify the account, contact, call-history, Safari, Calendar, Photos and KnowledgeC examinations. I also need to validate all timestamp conversions and finish the required supporting files. My final conclusions will remain limited to the evidence that can be supported by the extracted records and corroborating sources.

13. References

1. International Cybersecurity and Digital Forensics Academy, SBT-DF205 Case 2: iOS Forensics — iPhone SE iOS 13.4.1, supplied laboratory assignment.
2. International Cybersecurity and Digital Forensics Academy, supplied iOS forensic investigation slides.
3. Digital Corpora, Joshua Hickman iOS 13.4.1 iPhone SE evidence dataset, as identified in the laboratory assignment.
4. SQLite documentation, which was consulted to interpret database structures and SQL query behaviour.
5. Apple property-list and timestamp documentation, which was consulted to validate property-list formats and date representations.